IB

ECONOMICS

Poverty and inequality

P2 Questions

(Data Response)

QB6075

Name: …….

Grade: …….

Q1.

Text 1 - Budget 2022: Singapore unveils major tax measures to fund spending needs
Singapore on Friday (Feb 18) unveiled a slew of progressive tax measures aimed not only at generating revenues to fund major programmes needed over the next few years, but also at addressing social inequalities.

The hike in goods and services tax (GST) to fund the recurring social and healthcare needs of a rapidly ageing population was further delayed to 2023 in response to concerns over rising prices.

This year, the government will run up an expected overall **budget deficit** of $3 billion, amid a tone of cautious optimism sounded by Deputy prime minister Lawrence Wong as Singapore enters a period of transition and recovery after two years of grappling with the pandemic and its fallout.

Looking ahead, with government expenditures projected to increase significantly in the coming years - especially in healthcare - enhancements to Singapore's tax system would be needed to raise additional revenue, he added.

To that end, personal income tax will be increased from 2024. The portion of chargeable income in excess of $500,000 up to $1 million, will be taxed at 23 per cent, up from 22 per cent currently. Chargeable income in excess of $1 million will be taxed at 24 per cent.

Text 2 - The tale of two cities: Wage inequality in Singapore and the need for real productivity gains

The COVID-19 pandemic has peeled open the cracks of inequity in Singapore, dividing society across the lines of the “haves”—those who have the luxury of a skilled remote job or the financial cushion to weather the storm—and the “have-nots”—low-wage workers who cannot risk losing their jobs and essential workers on the frontline who cannot risk getting sick.

Despite government efforts at redistribution, the crux of the issue lies in the persistent trend in wage inequality. Between 2004 to 2014, growth in real incomes of households in the bottom 20%, before taxes and **transfer payments**, has been the slowest of any group, resulting in a 13.8% gap in growth rates between the lowest and highest quintile. Around 440,000 Singaporeans earning less than SGD$2,000 (US$1,415) per month fell into the bottom 20% of the resident workforce.

As a small open economy, Singapore faces the challenge of safeguarding its global competitiveness while keeping inflation at bay. Without substantial productivity gains, sectors in non-tradable industries—sectors that consist of locally-rendered services such as construction, retail and cleaning services—are unable to raise wages without leading to the negative ramifications of higher inflation.

Likewise, sectors in the tradable industries—sectors that serve the export markets such as manufacturing and wholesale trade—must be conscious of wage increases to retain competitiveness in the global market. The debate on wage inequality must keep in mind these real and pressing concerns around the unique vulnerabilities and constraints that small open economies like Singapore face.

Figure 1 – Gini coefficient data

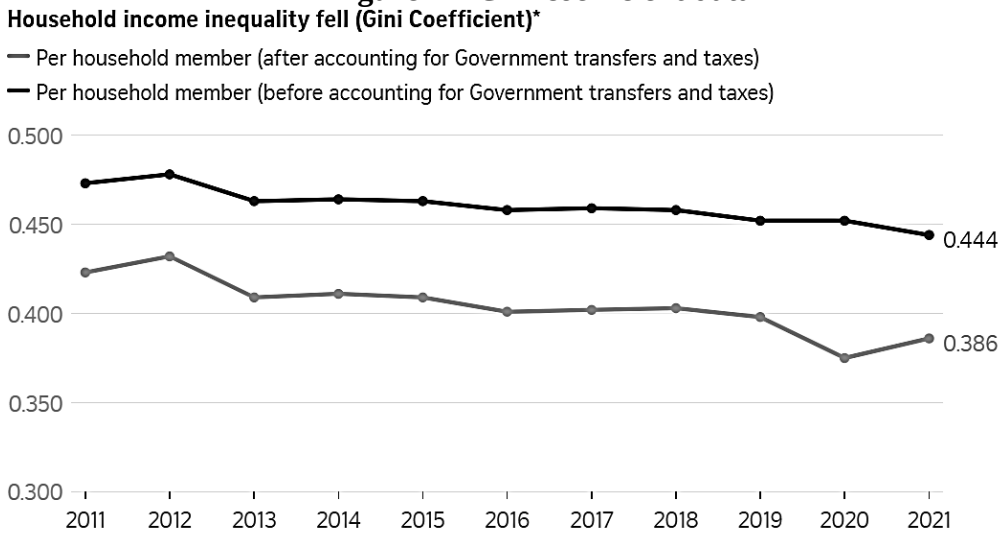


Figure 2 - Cumulative growth (%) of real average monthly income per member of household per quintile <u>before</u> taxes and transfer payments from 2004-2014

Quintiles	Cumulative growth (%)
1st (Poorest)	24.1
2nd	34.8
3rd	35.5
4th	36.2
5th (Richest)	37.9

Figure 3 - Cumulative growth (%) of real average monthly income per member of household per quintile <u>after</u> taxes and transfer payments from 2004-2014

Quintiles	Cumulative growth (%)
1st (Poorest)	45.7
2nd	40.5
3rd	34.3
4th	35.0
5th (Richest)	35.7

Q1.

(a) Define the following terms

(i) Budget deficit – Text 1 (2 Marks)

(ii) Transfer payments – Text 2 (2 Marks)

(b) Using a Lorenz curve, show the change in income inequality in Singapore before taxes and transfer payments from 2020 to 2021 – Figure 1 (3 Marks)

(c) Using an appropriate AD-AS model, explain the impact that a hike in the GST would have on economic growth and the price level in Singapore – Text 1 (4 Marks)

(d) Explain two negative consequences of the increase in the marginal income tax rate for highest income brackets in Singapore – Text 1 (4 Marks)

(e) Using a labour market diagram, explain why differential gains in labour productivity can lead to worsening income inequality – Text 2 (4 Marks)

(f) With reference to figures 2 & 3, analyse if taxes and transfers managed to reduce the income inequality between the highest and lowest quintiles from 2004 – 2009 (4 Marks)

(g) Using information from the extracts as well as your knowledge of economics, evaluate the impact of above-mentioned tax-reforms on income inequality in Singapore (15 Marks)

Q2.

Text 1 - Ranks of Mexican poor swell to reach nearly half the population

Mexico's poverty rate grew to nearly 44% at the end of last year, government data released on Thursday showed, worsened by an economic slump as the coronavirus pandemic led to steep budget cuts, business closures and layoffs.

Some 3.8 million more Mexicans fell below the official **poverty** line in 2020 to reach a total of nearly 56 million, compared with the end of 2018 when President Andres Manuel Lopez Obrador took office on a pledge to prioritize the needs of the poor.

Mexico's population of about 126 million is the largest among Spanish-speaking countries, second only to Brazil in Latin America.

Mexicans lacking access to health services grew 12 percentage points from 2018 to around 35 million people, marking the biggest jump among social deficiency indicators evaluated by the agency, the data showed. There was also
Mexico's economy contracted 8.5% last year as Lopez Obrador dismissed calls to take on new debt and stuck to his pre-pandemic policy of austere spending while boosting budgets for a range of social programs.

Text 2 – Colombia's leftist president presents ambitious tax plan

Colombia's first leftist president presented an ambitious tax plan on Monday that aims to raise up to $11 billion a year for anti-poverty programs.

Gustavo Petro was sworn into office on Sunday and has promised to fight economic inequality while investing in rural areas that have long been haunted by drug-related violence.

Less than 24 hours after taking his presidential oath, the new president presented a tax reform plan to his nation's congress that will increase income taxes on wealthy individuals and place a 10% levy on oil exports. The bill also includes an annual wealth tax on individuals who have a net worth of more than $750,000 as well as sales taxes on soft drinks and highly processed foods.

Tax collection in the South American Nation was worth roughly 19% of its total GDP in 2020, which is slightly below the Latin American average. Members of the OECD, an organization made up mostly of developed countries, collect an average of 33% of their GDP in taxes.

Governments in Colombia have long struggled with tax collection because much of the population works in the **informal economy**. Last year, an effort by conservative President Iván Duque to raise income taxes and some sales taxes sparked massive protests in which more than 50 people were killed.

Petro's incoming finance minister said the new tax reform plan will only attempt to raise income taxes on the wealthiest 2% of individuals. He has urged Colombia's wealthiest citizens to accept greater taxation and addressed the issue during his inauguration speech.

Text 3 - Peru poverty rate rises for first time in 16 years: government

Peru's poverty rate increased last year for the first time since 2001, rising one percentage point to 21.7 percent, according to government data published on Tuesday.

Peru had slashed its poverty rate by an average of about 4 percentage points between 2006 and 2012 on surging prices for the country's key mineral exports. But the decline slowed in recent years as the global commodities boom retreated.

Also in 2017, 375,000 Peruvians joined the ranks of the poor, which the state statistics agency Inei defines as an individual surviving on less than 338 soles ($105) per month.

Some 6.9 million Peruvians now live in poverty, 44 percent of whom are in rural Peru, Inei said. However, the rise in the poverty rate last year was led by a 2.3 percentage point spike in the capital Lima, home to some 10 million people.

"This is unacceptable. We must work harder to turn this around," Peru's President Martin Vizcarra told journalists in televised comments.

Vizcarra, a former vice president, replaced Kuczynski after he resigned last month and has vowed to refocus the government's efforts on infrastructure spending to help bolster economic growth that slowed to 2.5 percent last year.

In 2017, public investments contracted 2.8 percent and private investments grew 0.3 percent, according to central bank data.

Figure 1 – share of population living in poverty in Mexico between 2014 and 2020, by severity

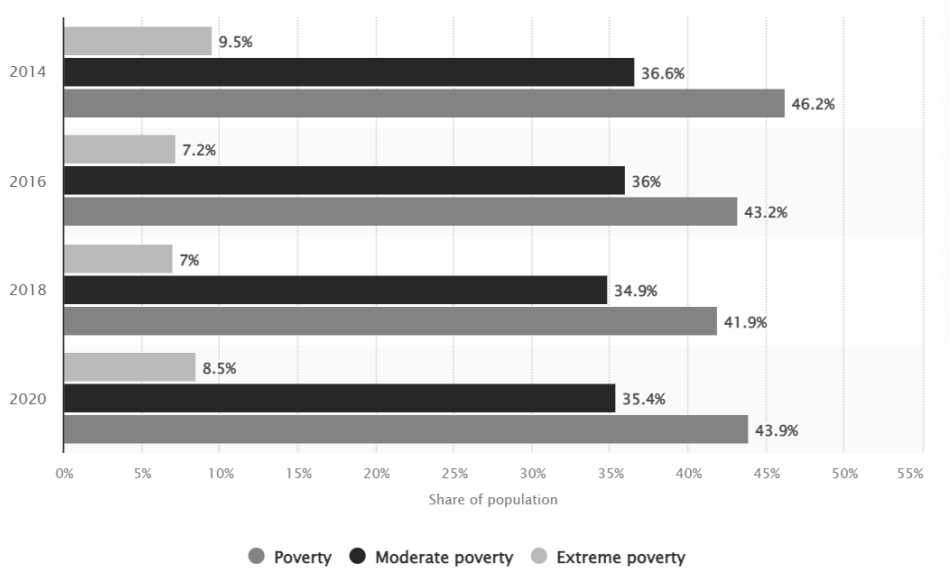


Figure 2 – Colombia's Gini index (Gini coefficient * 100)

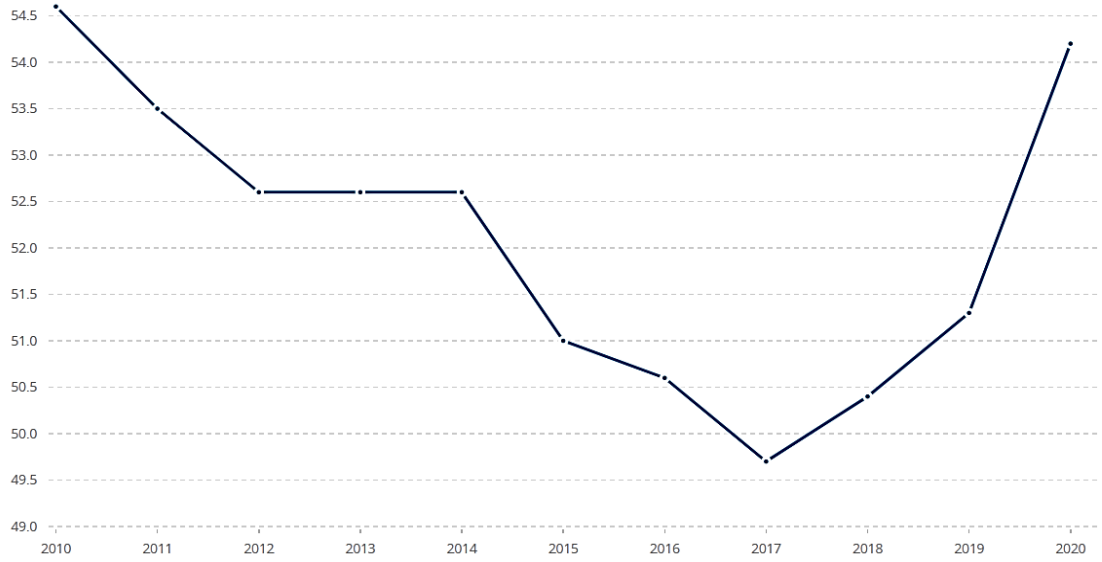
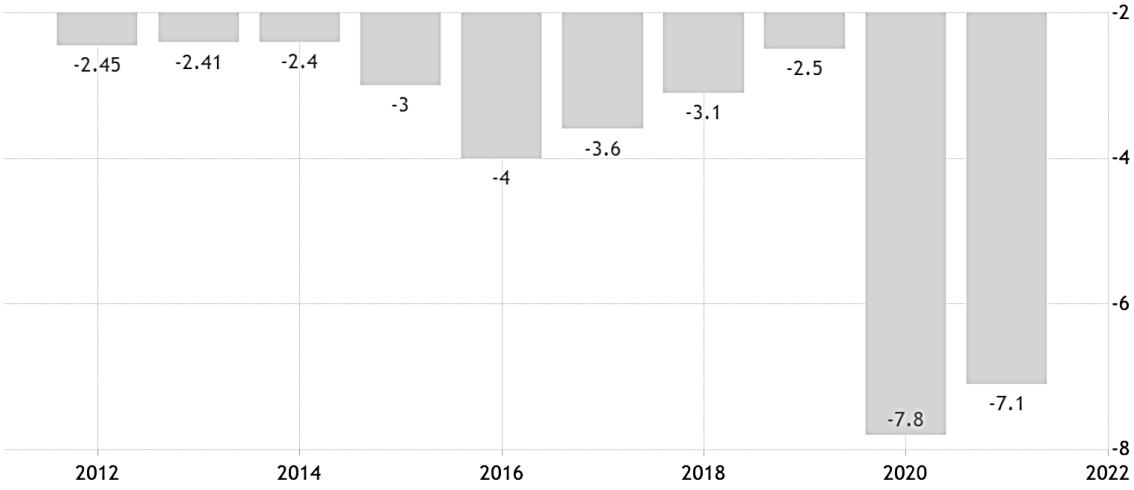


Figure 3 – Colombia's Government budget expressed as a percentage of GDP



Q2.

(a) Define the following terms:
    (i) Poverty (2 Marks)
    (ii) Informal economy (2 Marks)

(b) Using a business cycle diagram, identify where the Mexican economy could potentially be at – Text 1 (3 Marks)

(c) Using information from text 1, explain why income thresholds fail to account for the multi-dimensional nature of poverty. (4 Marks)

(d) Using a demand and supply diagram, explain a tax on oil exports can help low income households in Colombia to maintain their standards of living -Text 2 (4 Marks)

(e) Using a Lorenz curve and data from figure 2, explain how income inequality has changed in Colombian government since 2017 (4 Marks)

(f) Explain two disadvantages of raising income taxes on the wealthiest individuals – Text 2 (4 Marks)

(g) Using an AD-AS diagram, explain how rising mineral exports has helped reduce income inequality in Peru – Text 3 (4 Marks)

(h) Using information from the extracts as well as your knowledge of economics, evaluate if infrastructure spending in Peru is the most effective means of reducing poverty (15 Marks)

Q3.

Text 1 - Poverty Rising Again in Greece with Three Million at Risk

Poverty is on the rise in Greece with a total of 29.5 percent of the total population—around 3,092,300 people—at risk of poverty or social exclusion, according to an independent report published on Monday.

According to "Poverty Watch Greece 2022" of the Hellenic Antipoverty Network, which uses Hellenic Statistical Authority data, 19.6 percent of the Greek population is at risk of **income** poverty. Another 14.8 percent experiences material deprivation of basic goods, and a further 13.6 percent of eighteen to sixty-four-year-olds live in low-employment households.

"These figures show that child poverty in Greece is severe, recurrent," the report noted, "and worsening, and that we are in a much worse position than the European average, where 1 in 5 children live in poverty."

Relative poverty rates started to increase dramatically after 2010 (27.7 percent), starting to drop in 2015 (from 35.7 percent) and reaching 28.9 percent in 2020.

"It is striking that in the whole of the past twelve years, about a third of the total population survived in conditions of poverty or exclusion according to official data," the report says.

"This suggests that poverty has been a persistent problem for a large part of the population and that it only started to decline in the years 2017-2019," it was added.

In 2020, with the large drop in incomes due to the pandemic and its ensuing recession, relative poverty began to rise again.

Text 2 - Poverty in Greece Gone Up 40% Since 2008

The poverty in Greece increased by 40% from 2008 to 2015, according to a Cologne Institute of Economic Research study on European economy.
The study is not based on low income alone but on other factors as well, such as the deprivation of material goods, low quality education, underemployment and limited access to healthcare.

All the above factors are used in a new "**multidimensional poverty index**" according to which the Institute has conducted the survey for the years 2008-2015.

The Institute attributes the high level of poverty in Greece to years of recession and high unemployment, without excluding austerity policies imposed by lenders.

Text 3 – Why Greece needs Basic Income

Greece has a well-known problem with tax collection. According to OECD figures the country only succeeded in collecting just over half the tax revenue that it was owed in 2010. And even more potential revenue is lost through both professionals significantly underreporting their incomes, as Business Insider revealed.

With tax collection at such low levels, part of the problem must be both a deep distrust of state institutions by the public and a consensus that there will be few repercussions for tax evasion. Greece's former finance minister Yanis Varoufakis says one of the key aspects of the government's reform proposals is not only to clamp down on fraud and evasion but also, crucially, to "strengthen incentives to declare paid work".

This is where a Guaranteed Basic Income — a regular payment made directly by the state to people's bank accounts without the need to means check or include a work requirement — can help boost the state's revenue and reduce income inequality

This type of scheme does not suffer from the perverse incentives of existing means-tested welfare schemes that are withdrawn as recipients start to earn more. Means-tested welfare policies taking a low-paid job can result in people losing as much in state benefits as they gain in income from working. This drastically reduce the appeal of seeking work and can lead to what people call a "benefits trap". This would also bring more workers from the so-called "black economy" back into the tax system.

Text 4 – Opposition leader Alexis Tsipras compared Greek's current moment to the last days of Pompeii

Main opposition SYRIZA-Progressive Alliance leader Alexis Tsipras asked for general elections to be held immediately in his address on Tuesday. He accused the current Greek prime minister of incompetence, comparing the current period with "the last days of Pompeii".

Tsipras stressed the need for political change and a new plan to restore "political and economic stability". The pandemic and the Russian invasion of Ukraine have acted as a big bang for the international community and the global economy," Tsipras underlined, pointing out that most EU countries followed a different path to that of Greece, one which included "nationalisation in energy, increase in the minimum wage and activating mechanisms to control and restrain prices".

Tsipras warned of a new "debt crisis" that was again threatening the Greek economy, not of public but of private debt. He referred to a gigantic and unfair redistribution of income and of wealth from the many to the few and powerful, via high indirect taxation and profiteering in electricity rates".

Figure 1 – At risk of poverty rate in Greece

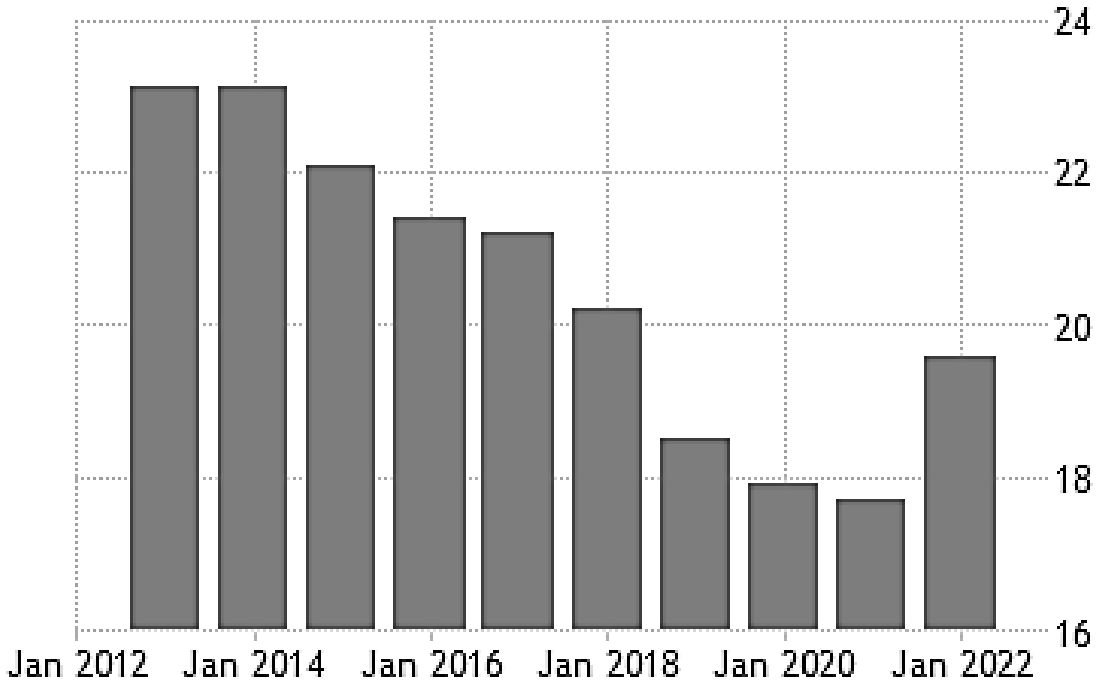
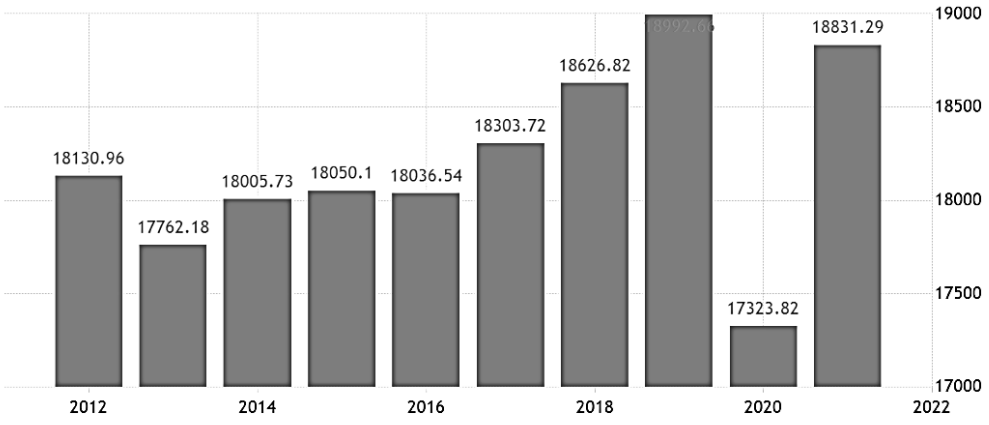


Figure 2 – RGDP per capita for Greece



Q3.

(a) Define the following terms:
(i) Income (2 Marks)
(ii) Multi-dimensional poverty index (2 Marks)

(b) Using figures 1 & 2, analyse the relationship between proportion of those at risk of poverty and RGDP per capita (3 Marks)

(c) Explain two reasons why the recession led to an increase in relative poverty – Text 1 (4 Marks)

(d) Explain why improved access to healthcare and education can contribute towards reducing both absolute and relative poverty – Text 2 (4 Marks)

(e) Explain two reasons why moving from a means-tested benefits system to one of universal basic income can help increase the government budget – Text 3 (4 Marks)

(f) Using a demand and supply diagram, explain how governments can use price controls to reduce the impact of rising energy prices on lower income households – Text 4 (4 Marks)

(g) Using a Lorenz curve, explain how higher indirect taxes will result in an increase in income inequality – Text 4 (4 Marks)

(h) Using information from the extracts as well as your knowledge of economics, evaluate universal basic income as a policy to reduce income inequality (15 Marks)

Markscheme

Q1.

(a)

(i) **Budget deficit:** Situation where /amount by which government expenditure exceeds government revenue

(ii) T**ransfer payments:** Payment made by the government to individuals for the purpose of redistributing income

(b) Using a Lorenz curve diagram, show a Lorenz curve shifting outwards due to an increase in income inequality as the Gini coefficient rises

(c) Using a monetarist AD-AS model, showing a leftward shift in SRAS, explain that:

- GST is an indirect tax that raises cost of production, reducing SRAS

- The fall in SRAS causes RGDP to fall, reducing economic growth

- The fall in SRAS increase price level, increasing inflation

(d) Explain that:

- Higher marginal tax rates could incentivize those who are most productive to leave country which could result in a net loss in tax revenue

- Higher marginal tax rates for higher income brackets reduce productivity as income gains from extra effort would be taxed at a higher rate

(e) Using a labour demand and supply diagram, showing two rightward shifts in labour demand at differing magnitudes, explain that:

- Different level of increase in labour productivity increases demand for labour to different degrees

- Resulting in difference in wage increase, worsening income inequality

(f) Explain that:

- Initiatives to redistribute income decreased income growth of the highest income quintile (i.e. 5th quintile) from 37.9% to 35.7%.

- These transfers raised income growth for the lowest quintile (i.e. 1st quintile), from 24.1% to 45.7%.

- Redistributive policies approximately doubled income growth for lowest quintile but lowered it for the highest quintile which reduces income inequality

(g) **Why tax reform worsens income-inequality:**

Increase in indirect tax (GST)

- Indirect taxes are regressive by nature as they take up a larger proportion of income of the poor relative to the rich

- Worsens existing cost-push inflation which disproportionately affect lower income households

- Reduction in RGDP from lower SRAS due to increase cost of production reduces RGDP and increase unemployment which disproportionately affects lower income households

Increase in direct taxes

- Direct taxes lower AD which reduce RGDP and increase unemployment which disproportionately affects lower income households

Why tax reform improves income-equality:

- Higher marginal tax rates on highest income tax brackets will reduce income inequality

- Revenue is being raised to make retirement and healthcare for the elderly more affordable which implies more transfers to relatively poorer, which decreases in income inequality

Q2.

(a)

(i) **Poverty:** The inability of an individual or household to obtain an adequate standard of living. This standard can be absolute or relative

(ii) **Informal Economy:** Part of the economy that lies outside the formal economy, consisting of economic activities that are unregistered and unregulated

(b) Using a business cycle diagram: Label a point in the downward sloping section of the business cycle below the trend line

(c) Explain that:

- Income thresholds assume that if individuals and households have access to a particular level of income, that necessarily guarantees them a particular level of access to goods and services

- But merit and public goods are underprovided in a market, so income does not necessarily guarantee access

(d) Using demand and supply diagram, show a rightward shift in supply in the domestic market for oil, explain that:

- Tax on exports raise costs of selling overseas, firms are more willing to sell domestically, increasing supply domestically

- Lowers price domestically which makes oil more affordable to lower income groups

(e) Using a Lorenz curve diagram, show a Lorenz curve shifting outwards, and explain that:

- Gini coefficient has risen, implying increase in income inequality

- This is represented by outward shift of Lorenz curve

(f) Possible disadvantages of higher income taxes on the rich:

- Brain drain resulting in reduction of productive capacity

- Loss in tax revenue due to brain drain

- Reduction in productivity due to less motivation to work harder

- Reduction in consumption leading to lower AD, reducing RGDP

(g) Using an AD-AS diagram, show a rightward in AD and explain that:

- Increase in exports will lead to an increase AD, increasing RGDP

- RGDP increases implies income increases and lower unemployment, which could increase lower income group's investment towards human capital (e.g. Education)

- Increases their income growth in the long term thus narrowing the income disparity

(h)

(i) **Infrastructure spending is the most effective means of reducing poverty**

- Increases in government spending will increase AD, increasing economic growth and reduce unemployment, reducing absolute poverty

- Investment towards infrastructure might attract inward Foreign Direct Investments (FDI) which further create job opportunities when multi-national corporations (MNCs) hire locals

- Infrastructure spending can be targeted at improving access to low income individuals to merit and public goods (Multi-dimensional poverty)

- Increases in productive capacity ensures sustainable increases in income in the long run

(ii) **Infrastructure spending could worsen poverty**

- Tax increases might be necessary to pay for infrastructure spending which might be raised in a regressive manner

- Opportunity cost of the spending (could target lower income individuals directly reducing absolute poverty and relative poverty via transfer payments instead)

-Investment of infrastructure in urban areas alone might not effectively alleviate poverty rates in rural areas as rural inhabitants have limited benefits

-Unequal investment in infrastructure might also create more inhabitants to relocate in urban areas which drive cost of living in the area to be higher thus resulting in urban poverty

(iii) **Alternative policies to reduce poverty**

- Increases in minimum wages could be costless means of reducing absolute and relative poverty

- As mentioned above, transfer payments could also reduce absolute and relative poverty more directly

Q3.

(a)

(i) **Income:** money earned by a household from the supply of a factor of production

(ii) **Multi-dimensional poverty index:** A composite indicator that measures poverty in the three dimensions: health, education and living standards

(b) Possible comments:

- No relationship between RGDP per capita and proportion at risk of poverty from 2014 -2016

- 2016-2019 suggests inverse relationship between RGDP per capita and proportion at risk of poverty

- Economic reasoning suggests increases in RGDP per capita implies income increases leading to greater access to goods and services reducing poverty risk

- 2020 is an anomaly where proportion at risk falls even though RGDP per capita falls significantly

(c) Possible reasons include:

- Reduces in RGDP implies falling incomes nationally, since low wage workers tend to be unskilled, they have limited bargaining power resulting in their income decreases by disproportionately larger amount

- Unemployment increases as RGDP falls, low-income individuals are disproportionately less skilled and have limited bargaining power making them more likely become unemployed.

- Unemployment increases as RGDP falls resulting in high levels of inequality in income as the rich might have alternative source of income unlike the poor thus increasing relative poverty.

(d) Possible reasons include:

- Government provision of education and healthcare increased AD, resulting in higher RGDP which implies increases in income (or reduced unemployment), reducing absolute poverty

- Increased access to education and healthcare improves the relative productivity of workers from lower income households, causing their wages to relatively increase, reducing relative poverty

(e) Possible reasons why:

- More incentive to declare paid work and pay taxes on it as will not be prevented from earning basic income as there is no income threshold

- More incentive to work harder and earn more as there is no income threshold, which increases the tax revenue paid

(f) Using a demand and supply diagram, show a price ceiling (below equilibrium price), and explain that:

-Price ceiling makes energy more affordable to lower income households, leaving a larger proportion of income to be spent on essential goods

-Energy is an input in production of making goods and services, and lower energy prices keep the price of final goods relatively lower

(g) Using a Lorenz curve diagram, show an outward shift of the Lorenz curve and explain that:

- Indirect taxes are regressive by nature as all individuals pay the same increase in price as tax regardless of income

- This worsens income inequality as the poor pay a larger proportion of income as tax, which causing the Lorenz curve to shift outwards

(h) **Advantages of universal basic income:**

- Could increase incomes relatively more for lower income households as they have greater access to education and healthcare increasing their relative productivity

- Direct impact on income inequality as it is a transfer payments

- More choice in how money is spent by the individuals that receive it

Disadvantages of universal basic income:

- Costly which implies a strain on government budget

- Might require an increase in direct taxes (if progressive) which might reduce motivation for high skilled workers

- Could cause demand-pull inflation as transfers might lead to more consumption and increased AD

- Opportunity cost of government expenditure